Example 1.13 Let $A=\{1,2,3\}$, $B=\{4,5,6,7\}$ and $f=\{(1,4),(2,5),(3,6)\}$ be a function from A to B. Show that f is one – one but not onto function.

Solution $A=\{1,2,3\}$, $B=\{4,5,6,7\}$; $f=\{(1,4),(2,5),(3,6)\}$

Then f is a function from A to B and for different elements in A, there are different images in B. Hence f is one–one function. Note that the element 7 in the co-domain does not have any pre-image in the domain. Hence f is not onto (Fig.1.36).

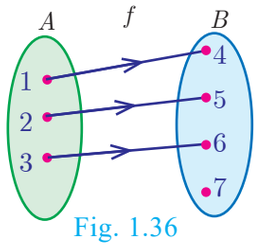


Fig. 1.36

$\therefore$ f is one–one but not an onto function.

Example 1.14 If $A=\{-2,-1,0,1,2\}$ and $f:A\to B$ is an onto function defined by $f(x)=x^2+x+1$ then find B.

Solution Given $A=\{-2,-1,0,1,2\}$ and $f(x)=x^2+x+1$.

$$\begin{aligned} f(-2) &= (-2)^2+(-2)+1=3;\\ f(-1) &= (-1)^2+(-1)+1=1\\ f(0) &= 0^2+0+1=1;\\ f(1) &= 1^2+1+1=3\\ f(2) &= 2^2+2+1=7\\ \therefore\ B &= \{1,\ 3,\ 7\}. \end{aligned}$$

Example 1.15 Let f be a function $f:\mathbb{N}\to\mathbb{N}$ be defined by $f(x)=3x+2, x\in\mathbb{N}$

(i) Find the images of 1, 2, 3 (ii) Find the pre-images of 29, 53

(ii) Identify the type of function

Solution The function $f:\mathbb{N}\to\mathbb{N}$ is defined by $f(x)=3x+2$

(i) If $x=1$, $f(1)=3(1)+2=5$

If $x=2$, $f(2)=3(2)+2=8$

If $x=3$, $f(3)=3(3)+2=11$

The images of 1, 2, 3 are 5, 8, 11 respectively.

(ii) If x is the pre-image of 29, then $f(x)=29$. Hence $3x+2=29$

$3x=27\Rightarrow x=9$.

Similarly, if x is the pre-image of 53, then $f(x)=53$. Hence $3x+2=53$

$3x=51\Rightarrow x=17$.

Thus the pre-images of 29 and 53 are 9 and 17 respectively.

(iii) Since different elements of $\mathbb{N}$ have different images in the co-domain, the function f is one – one function.

The co-domain of f is $\mathbb{N}$.

But the range of f = {5, 8, 11, 14, 17, ...} is a proper subset of $\mathbb{N}$.

$\therefore$ f is not an onto function. That is, f is an into function.

Thus f is one – one and into function.

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